

1. INTRODUCTION

The project is a simple wooden electrical circuit protection box designed as a protective enclosure for switches, small electrical circuits, or control components.

Unlike an ordinary storage box, this model does not have a closed base. The open side allows the enclosure to be positioned against a mounting surface or around an electrical component arrangement. A hinged cover is provided at the top, allowing easy access for inspection and maintenance.

A metal latch is fitted at the front to keep the cover securely closed. Metal angle brackets are also provided for additional support and stability.

The project demonstrates basic engineering workshop operations such as measuring, marking, cutting, drilling, sanding, fastening and fitting of mechanical components.

2. MATERIALS USED

- * Wooden boards / plywood pieces
- * Wood glue
- * Small wood screws
- * Metal hinge
- * Metal latch
- * Metal L-brackets / angle brackets
- * Sandpaper
- * Wood finish / paint or varnish, if required

3. TOOLS AND EQUIPMENTS

- * Steel ruler / measuring scale
- * Pencil
- * Hand saw / wood cutting saw
- * Drill machine
- * Suitable drill bits
- * Screwdriver
- * Clamps
- * Sandpaper / sanding block

4. DESIGN DESCRIPTION

The model is a rectangular wooden enclosure formed using wooden panels. The front portion is provided with a hinged cover that can be opened when access to the components inside is required.

A metal latch is fixed to the front of the enclosure. When the cover is closed, the latch holds it in position.

The bottom or rear mounting side is intentionally left open instead of using a solid base. This allows the enclosure to be mounted around a switch, control circuit or other component arrangement.

Metal angle brackets are fixed near the lower edges to provide additional support and stability.

The simple construction makes the model suitable for demonstrating the basic concept of an electrical or control-component enclosure.

5. MANUFACTURING / ENGINEERING PROCESS

5.1 Measuring and Marking

The required dimensions of the wooden panels are measured using a steel ruler. The cutting lines are marked using a pencil and try square.

Accurate marking is important to obtain properly aligned panels and a good final fit.

5.2 Cutting

The wooden boards are cut according to the marked dimensions using a hand saw or suitable wood-cutting tool.

Care is taken to maintain straight and square edges.

5.3 Drilling

Pilot holes are drilled at the required locations for mounting the hinge, latch and metal brackets.

Drilling helps in proper screw fastening and reduces the possibility of splitting the wooden material.

Additional holes can also be provided for cable or component access where required.

5.4 Sanding

The cut edges and surfaces are sanded using sandpaper.

Sanding removes rough surfaces, sharp edges and small irregularities and gives the model a smoother appearance.

5.5 Assembly

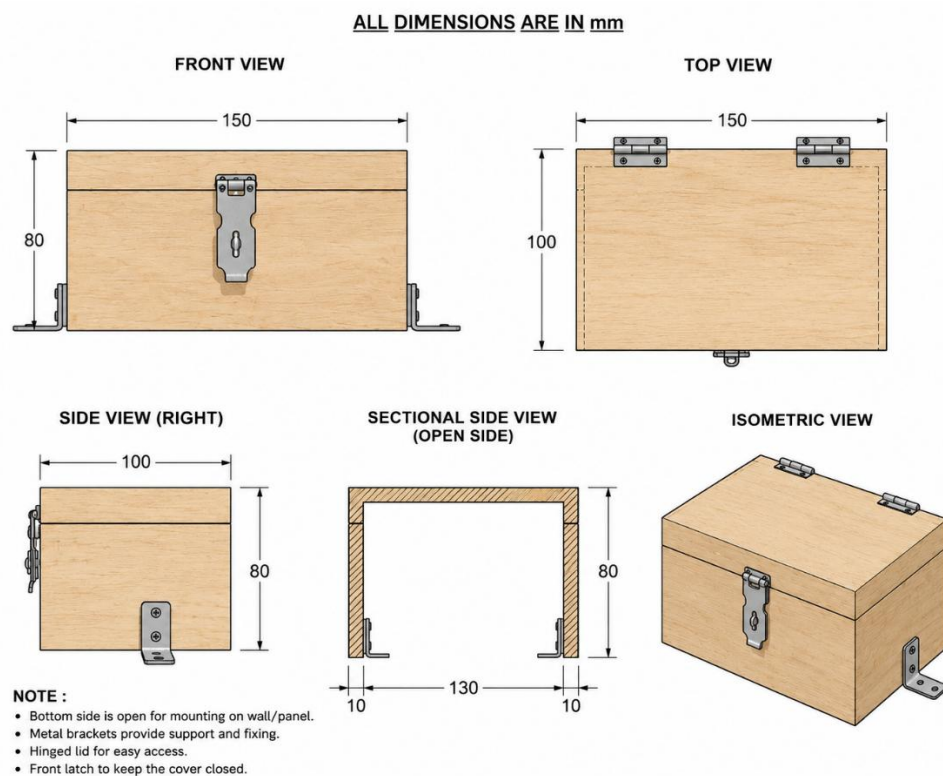
The wooden panels are positioned and aligned to form the enclosure.

Wood glue and/or screws are used to join the panels. The open mounting side is retained instead of fitting a solid base.

5.6 Hinge and Latch Fitting

The hinge is fixed between the main enclosure and the cover.

The front latch is then positioned so that the cover can close properly and remain secured.



5.7 Final Inspection

The completed model is inspected for:

- * Proper alignment of wooden panels
- * Smooth movement of the hinged cover
- * Proper operation of the latch
- * Tightness of screws
- * Stability of the metal brackets
- * Smoothness of the wooden surfaces

6. WORKING

The wooden enclosure acts as a protective cover around a switch, small electrical circuit or control component.

The hinged cover can be opened when access or inspection is required. After the component arrangement is placed inside or behind the enclosure, the cover can be closed.

The front latch holds the cover in the closed position.

The open mounting side allows the enclosure to be positioned against a suitable mounting surface while the wooden walls and cover provide physical protection around the component area.

7. APPLICATIONS

The enclosure can be used as an educational prototype for:

- * Protecting switches and switch assemblies
- * Covering small low-voltage electrical circuits
- * Housing simple control or demonstration circuits
- * Protecting small electronic/control components
- * Demonstrating basic electrical enclosure design

8. OBSERVATIONS

1. Accurate measurement and marking are necessary for proper panel alignment.
2. Proper cutting produces better joints between the wooden panels.
3. Drilling pilot holes improves screw fastening.
4. Sanding improves the safety and appearance of the model.
5. Correct hinge alignment is necessary for smooth opening and closing.
6. The latch keeps the cover securely closed.
7. The open mounting side makes the model suitable for mounting around a component arrangement.
8. Metal brackets provide additional support and stability.

9. SAFETY NOTE

This project is an **educational prototype** and should not be treated as a certified enclosure for mains-voltage electrical systems.

Wood is not a substitute for a properly rated electrical enclosure. For actual mains-voltage applications, suitable insulated, flame-resistant and certified enclosure materials must be used along with appropriate electrical protection.

10. CONCLUSION

The wooden electrical circuit protection box is a simple engineering model demonstrating the basic concept of an enclosure with a hinged access cover.

The project provided practical experience in measuring, marking, cutting, drilling, sanding, fastening and fitting mechanical hardware.

The open mounting side, hinged cover, front latch and supporting brackets make the model suitable as an educational prototype for demonstrating the protection and accessibility of switches, low-voltage circuits and control components.

The project also helped in understanding how basic workshop operations can be combined to produce a useful engineering product.

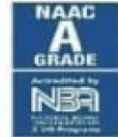




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PROJECT REPORT

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